

Keywords: Red blood cell, white blood cells, platelets, plasma, haemoglobin, medium

Starter: Complete these quiz questions about last lesson

TIME FOR A

QUIZ!



Quiz: Food Tests

1. What is the test for protein a) Biruet b) iodine c) ethanol d) benedicts
2. What is the test for starch?
3. What is the test for fats?
4. What is the test for sugar?
5. In the benedicts test the test tubes have to be placed into what for the substance to react
6. A positive result for the biuret test is a) blue/black b) yellow c) purple
7. The positive result for starch is
8. What two factors can cause enzymes to denature
9. Enzyme + sub_____ causes a reaction. Enzymes speed things up in living things so they are called B_____ C_____
10. Match the enzyme to the function

A) Amylase	A) Breaks down fats
B) Pepsin	B) Breaks down carbonhydrates
C) Lipase	C) Breaks down protein

The Blood

08/06/2026

Keywords: Red blood cell, white blood cells, platelets, plasma, haemoglobin, medium

Starter: Answer these unusual questions.

EXT: Write down some things you already know about blood.

1. Is the colour of our blood sometimes blue?



2. Does the amount of blood we have change if we did exercise?



3. True or false: It is our blood that helps make a scab to heal a wound.



4. Do we have to be careful when giving blood to other people? Why or why?



Review

1. Is the colour of our blood sometimes blue? **No! It is never blue.**



2. Does the amount of blood we have change if we did exercise? **Yes! our blood volume increases.**



3. True or false: It is our blood that helps make a scab to heal a wound. **TRUE!**



4. Do we have to be careful when giving blood to other people? Why or why?

Yes! There are different blood groups. You can only receive blood from the same group. Also we need to make sure the blood is free of pathogens and abnormalities



The Blood

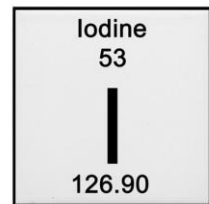
08/06/2026

Keywords: atrium, ventricles, valves, pump, chambers, cardiac muscle, coronary arteries, veins, aorta,

Starter: Answer these three review questions below.

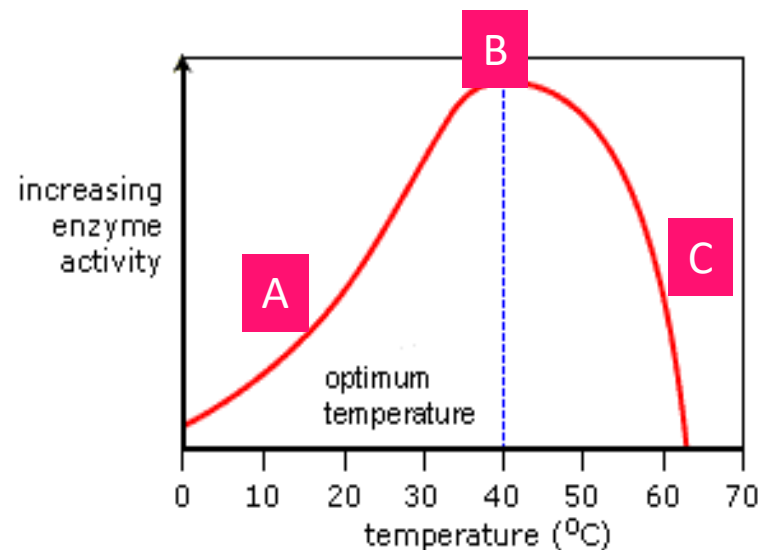
EXT: How are our bodies organised?

1. What colour does iodine turn when starch is present? [1 mark]



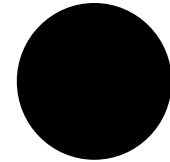
2. Enzymes can denature. When? Why? What happens to rate of the reaction? [3]

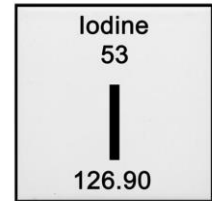
3. Explain A, B and C on the graph. [4]



Food Tests

1. What colour does iodine turn when starch is present?

 **Black**



2. Enzymes can denature.
When? Why? What happens to rate of the reaction?

When: acidic or alkaline pH or high Temperatures [1]

Why: Damage to the structure of Proteins [1]

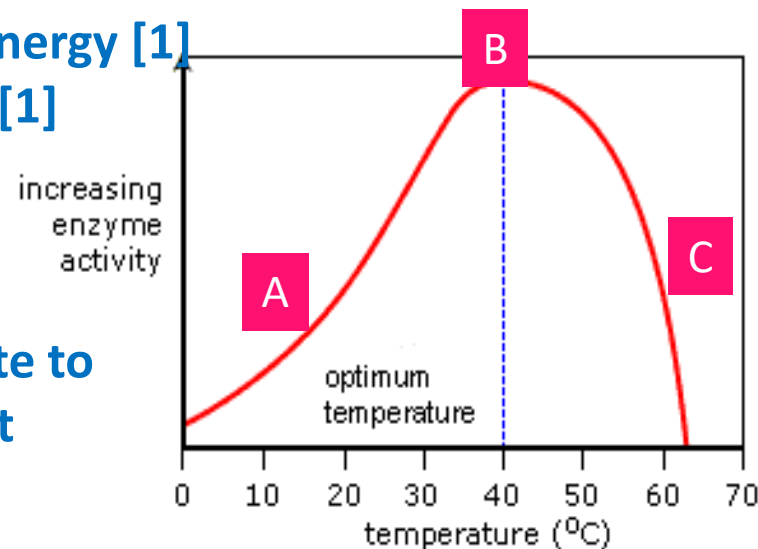
Rate of reaction will decrease [1]

3. Explain A, B and C on the graph.

A Particles have low kinetic energy [1]
low frequency of collisions [1]

B Optimal temperature [1]

C Denaturing causes active site to change shape, enzyme can't catalyse a reaction. [1]



LOb: Understand the components of the blood.

Keywords: red blood cell, white blood cells, platelets, plasma, haemoglobin, medium

Learning Outcomes:

- Recall substances transported by the blood to and from cells
- Recall and describe the main components of the blood and explain their functions



Did you know... 7% of the average persons weight is made up of their blood. The average person has about 5 litres of blood which is about 2 ½ coke bottles full. A human would die if they lost half of the blood that they have.

The function of the blood is to **carry dissolved substances** around the body that are **needed by the cells** of our body.

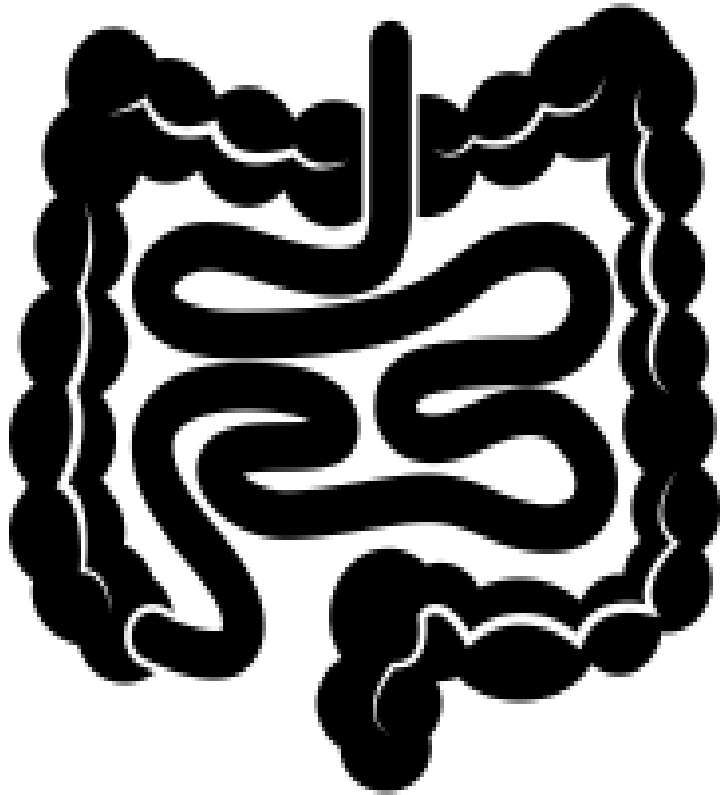


Oxygen (O_2)
Carbon Dioxide
(CO_2)

1 Gases

The function of the blood is to **carry dissolved substances** around the body that are **needed by the cells** of our body.

Glucose and
ions



2

Digested
molecules

The function of the blood is to **carry dissolved substances** around the body that are **needed by the cells** of our body.

Found in urine, urea is formed by proteins breaking down in the liver.
It is toxic.

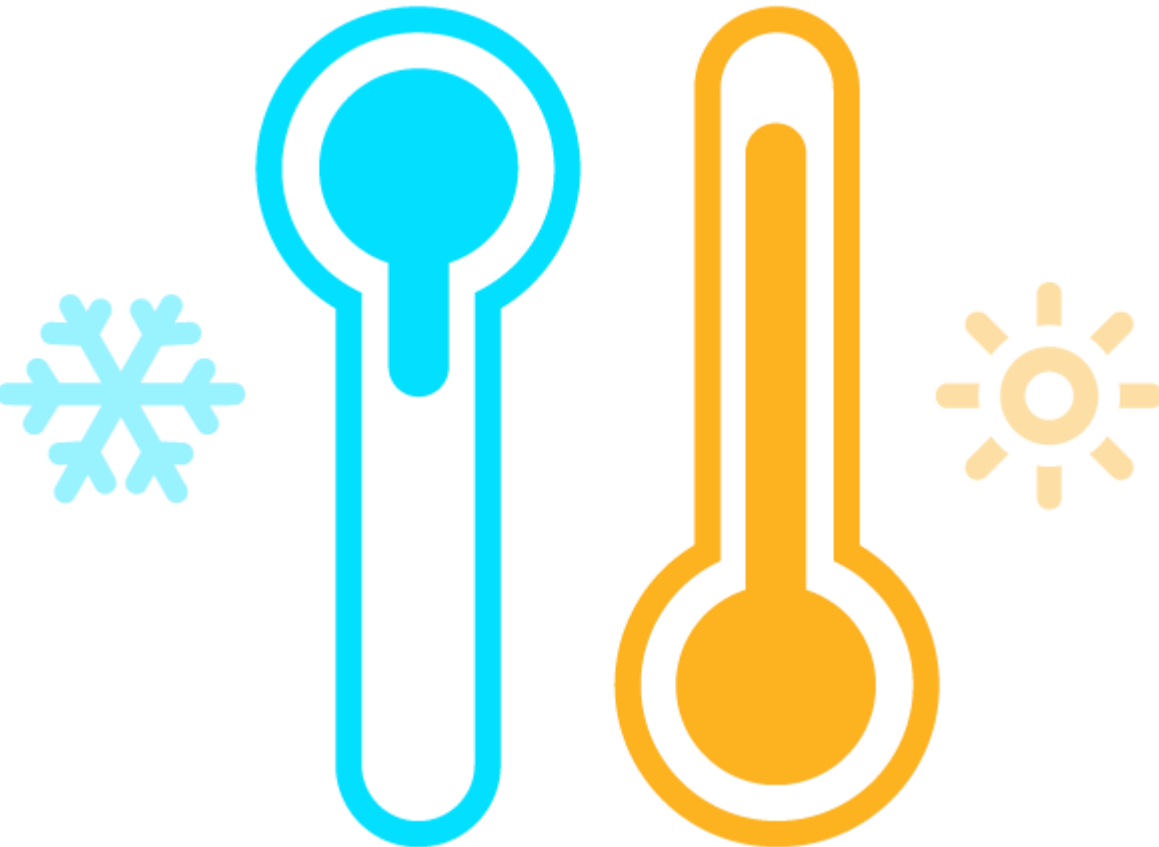


3 Urea

Another function of the blood is **regulate temperature**
(called **homeostasis**).



Regulate
Temperature



Activity: Two students are going to summarise in 30 seconds what we have learn so far. Then a student will become the teacher and ask questions to random members of the class.

30

**WALK IN
MY SHOES**

LOb: Understand the components of the blood.

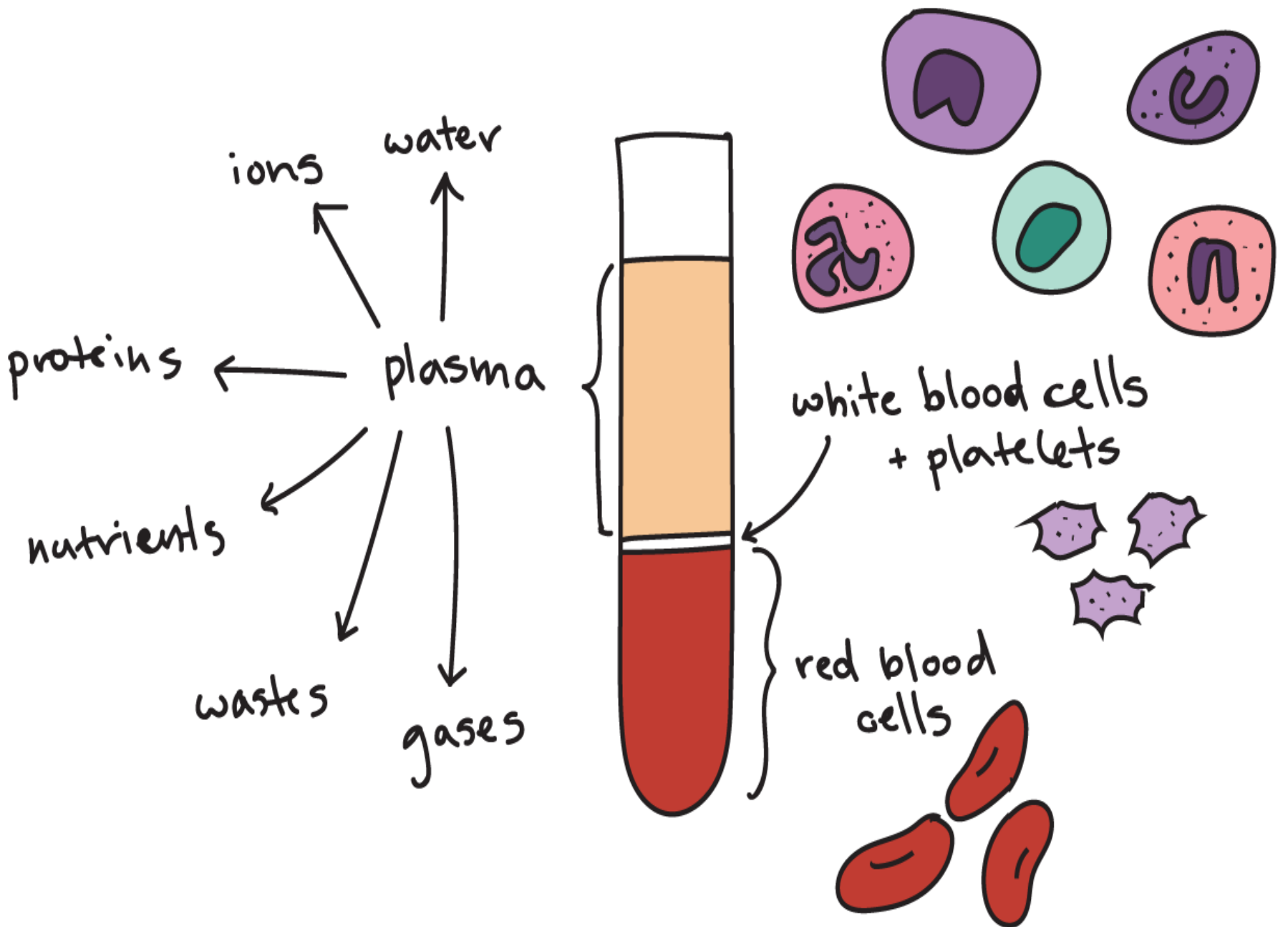
Keywords: red blood cell, white blood cells, platelets, plasma, haemoglobin, medium

Learning Outcomes:

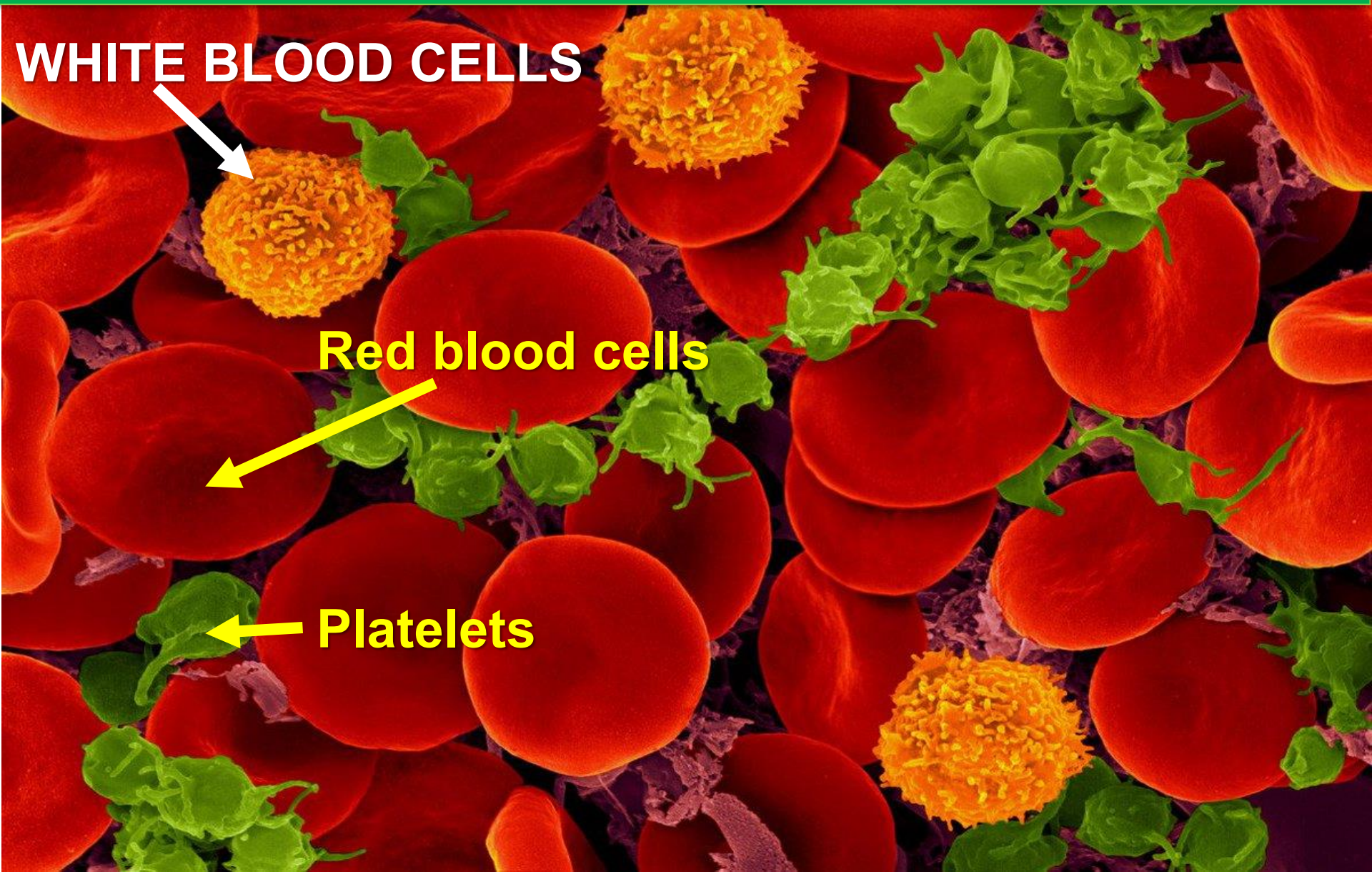
- ✓ Recall substances transported by the blood to and from cells
- Recall and describe the main components of the blood and explain their functions



Did you know... 7% of the average persons weight is made up of their blood. The average person has about 5 litres of blood which is about 2 ½ coke bottles full. A human would die if they lost half of the blood that they have.



There are four components to blood. What's the missing one?



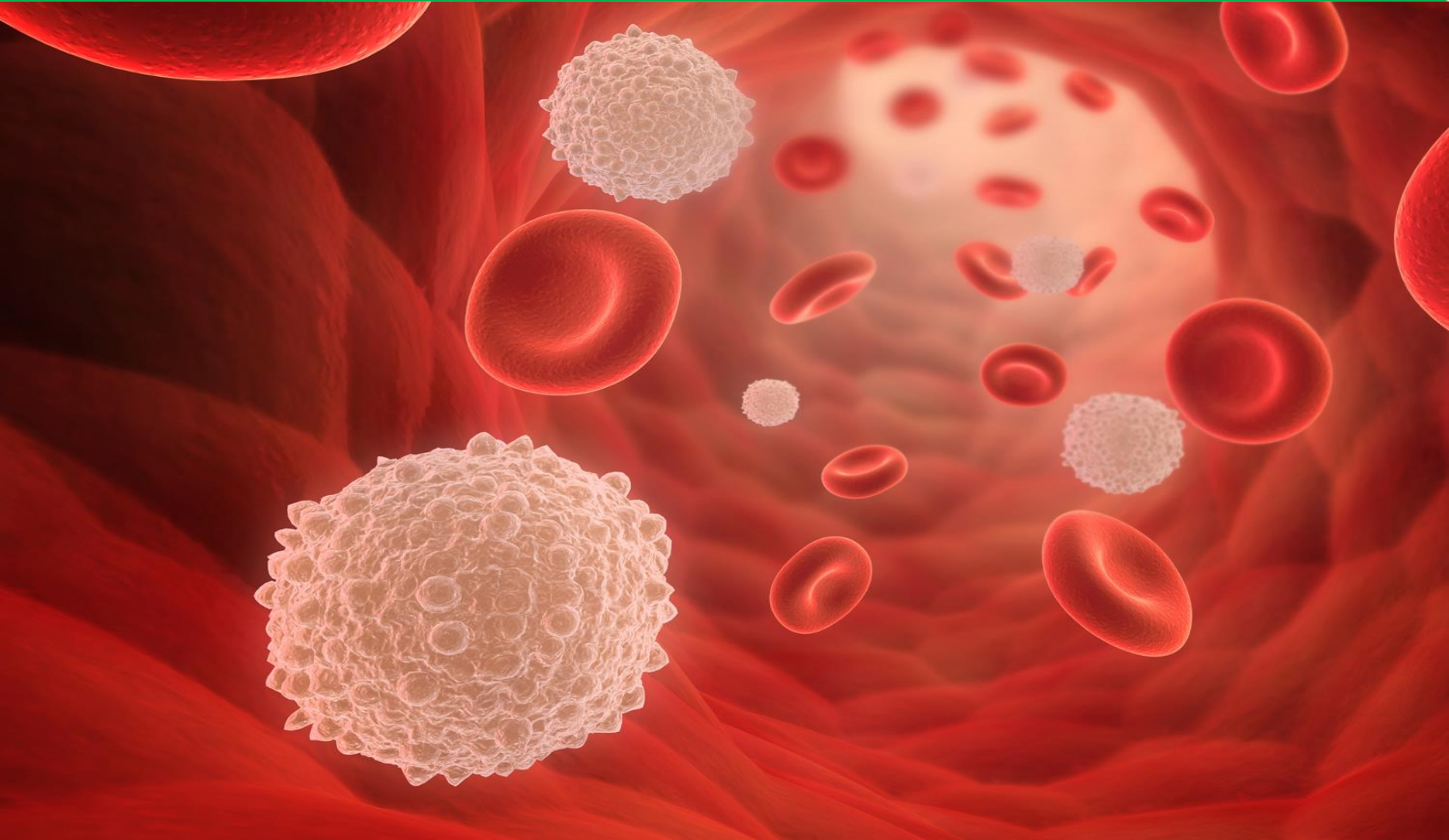
All the cells in the blood sit in a fluid called **PLASMA**.
It is yellow in colour. It is what the dissolved
substances are carried in.



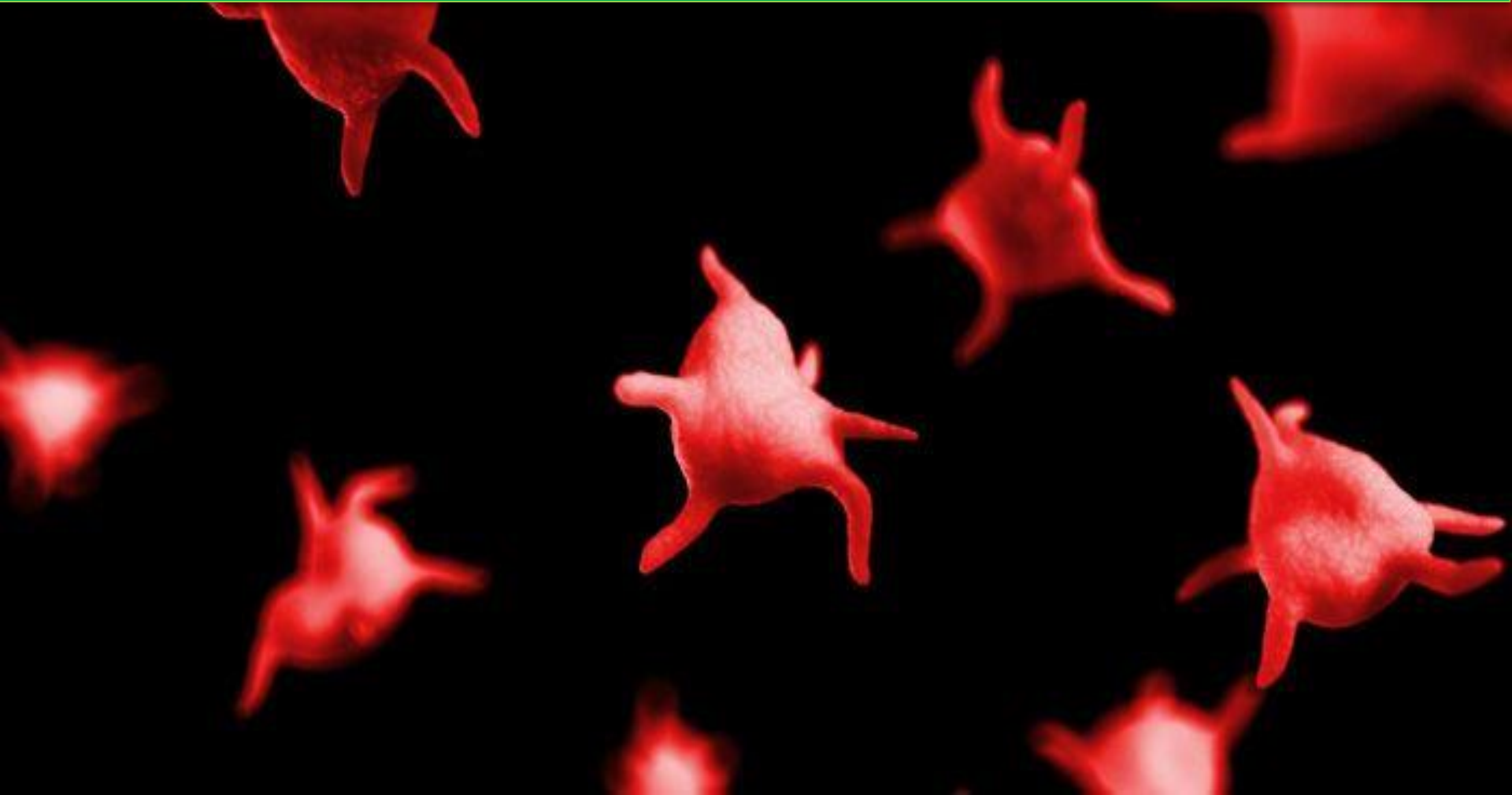
Red Blood Cells: Carry oxygen to the cells. They contain lots of molecules called **haemoglobin**. When oxygen binds to the haemoglobin we call it **oxy-haemoglobin**.



White blood cells: Larger than red blood cells. They are involved in fighting diseases. They can move freely from the blood into the tissue.



Platelets: Tiny cells that help block a wound and form a scab to stop bleeding. When we start to bleed a signal is sent that makes the platelets rush to the wound and block it up.



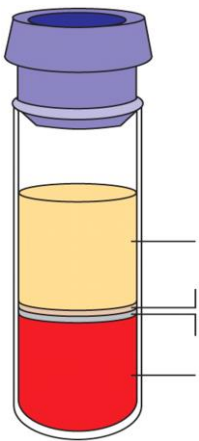
Activity: Use the information on your worksheet to complete the questions and the picture board.



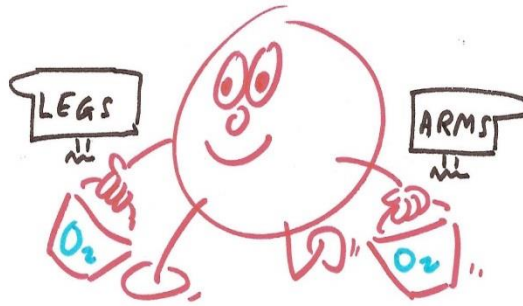
Activity: Use the information on your worksheet to complete a the questions and the picture board.

EXT: Use the text book to add extra information around your sheet make note of the key words.

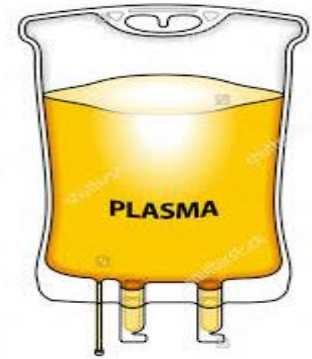
		
Blood is a mixture of...		
		 CLOT



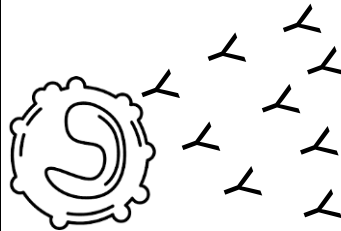
The purpose of blood is to...



The function of plasma is to...



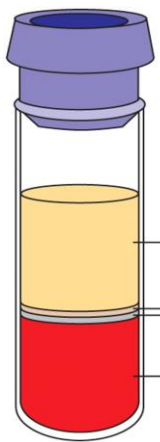
The first white blood cell is called a _____. It's function is to...



The second is called a _____.



CLOT

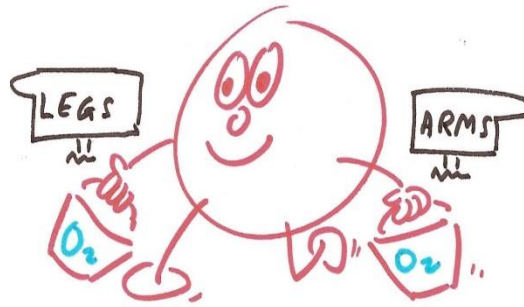


55% Plasma

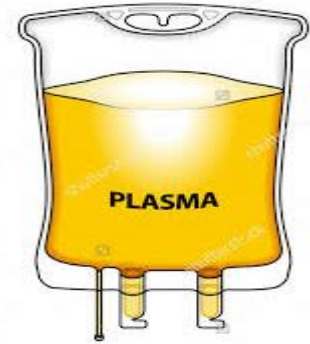
1% WBC, platelets

44% Red blood cells

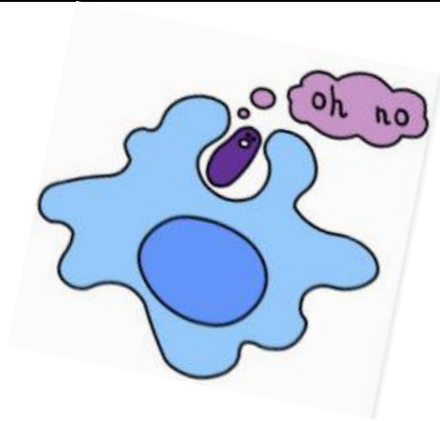
The purpose of blood is to...



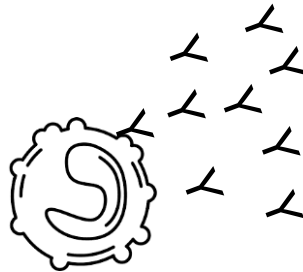
_____ transport
_____ and carbon dioxide.
This O₂ gas is carried by
_____ molecules and
forms oxy-_____



Plasma carries h_____
and small soluble
substances, such as
g_____



There are two types of
_____.
Phagocytes engulf

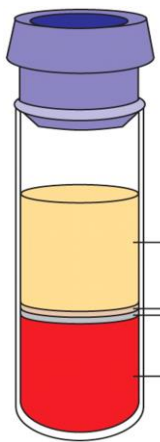


WBC called lymphocytes
produce_____.
These stick to the
_____ stopping them



CLOT

P_____ stop you
from _____. They
rush to the wound and
block it forming a clot.

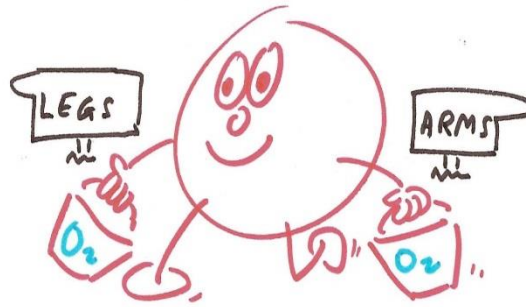


55% Plasma

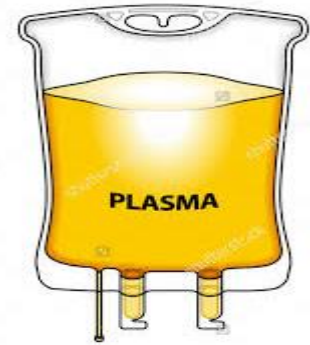
1% WBC, platelets

44% Red blood cells

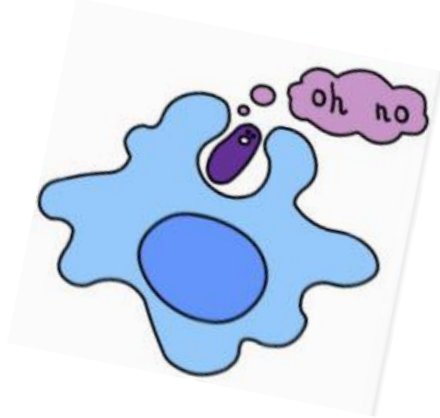
The purpose of blood is to...
carry dissolved substances
to the cells of the body



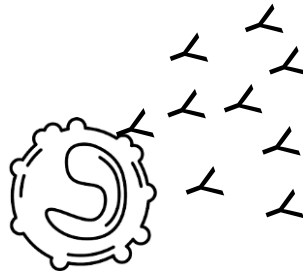
Red blood cells transport
oxygen and carbon dioxide.
Oxygen is carried by
haemoglobin and forms
Oxy-haemoglobin.



Plasma carries hormones
and small soluble
substances, such as
glucose



There are two types of
white blood cell.
Phagocytes engulf
pathogens



WBC called lymphocytes
produce antibodies. These
stick to the pathogen
stopping them moving.



CLOT

Platelets stop you from
bleeding. They rush to
the wound and block it
forming a clot.

Review

1. If you had to get rid of one component of the blood what would it be and why?

Get rid of platelets. There are people who have no ability to clot their blood and they can survive if they are careful. You couldn't survive long without the other 3 components.

2. By what process does oxygen get into and out of the blood? Explain.

☒ Diffusion ☐ Haemoglobin ☐ Osmosis

Molecules move from a high concentration to a low concentration from the lungs to the tissues.

3. Describe how the red blood cell is adapted to carry oxygen

☒ It has no nucleus so it is able to carry more oxygen

☒ It has a special molecule called haemoglobin

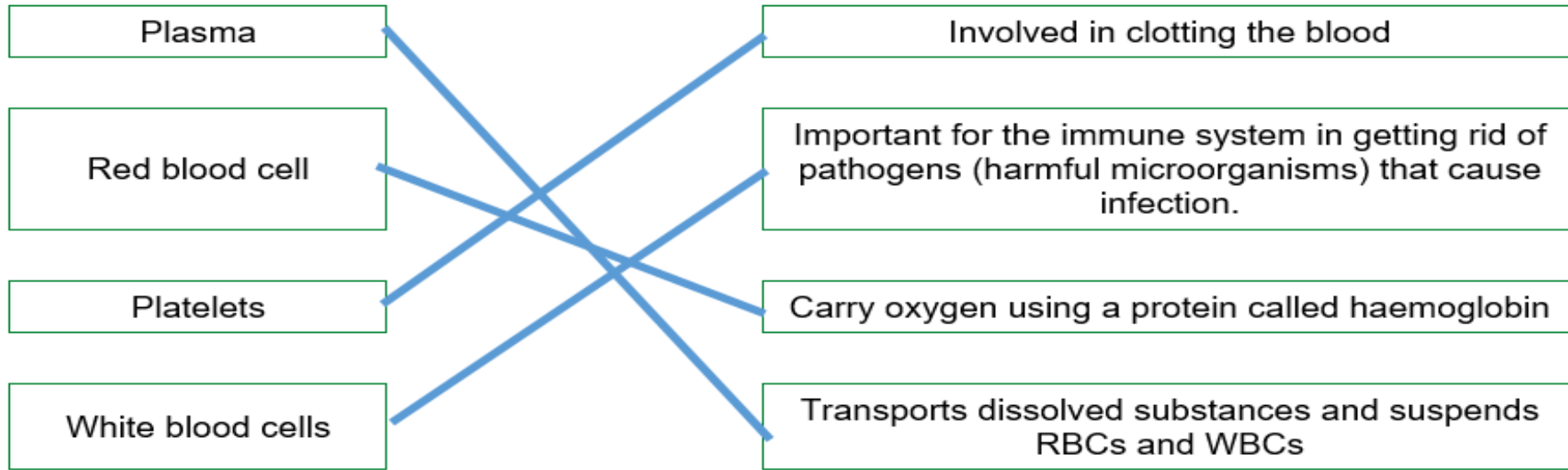
☐ It is red

☐ ~~It has a small surface area~~ Bio concave shape for a large surface area

Review

Questions

1 Link each organ with its function in the digestive system:



(3 marks)

a State a function of lymphocytes and explain how they are adapted to their function.

Function: Get rid of harmful microorganisms like pathogens. [1]

Adaptation: They can produce and release antibodies that bind with pathogens to destroy them. [1]

(2 mark)

b Explain how two adaptations of RBCs help them perform their function.

They have no nucleus. This allows more haemoglobin into the cell so that it can carry more oxygen [1]

They are bio-concave which increases their surface area allow more oxygen to be carried [1]

2

- a Name a soluble nutrient dissolved in plasma used for respiration.

Glucose

(1 mark)

- b What is a waste gas, made during respiration, that is released by the lungs?

Carbon dioxide

(1 mark)

- c Why is it useful that the reaction of haemoglobin and O_2 is reversible?

Allows red blood cells to pick up oxygen at the lungs and release it when it reaches the cells.

(2 mark)

- d What is phagocytosis ?

A process where phagocytes engulf pathogens to destroy them.

2

- e Describe what would happen if there was no plasma?

Other components could not be suspended and they could not be transported to other parts of the body.

[1]

Couldn't dissolve substances like glucose or oxygen

Respiration wouldn't happen we would die [1]

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key facts



There are four components. **Plasma, red blood cells, white blood cells** and **platelets**.



Plasma carries **soluble substances** like **glucose, urea, ions** and **hormones** and allows them to be **transported**.



Red blood cells carry **oxygen**. Three important adaptations. Contain **haemoglobin protein** to **carry oxygen**, no nucleus to pack in more haemoglobin and **bio-concave shape** to **increase surface area**.



White blood cells are part of the **immune system** and **fight infections**. **Phagocytes** are the most common. They **engulf** and destroy pathogens by a process called **phagocytosis**. **Lymphocytes** release **antibodies** that bind pathogens.

Plenary: Describe how each picture relates to today's lesson.

